TAlpha Nuclide Database Format Specification

Specification version 3.0

Property	Value
Format	TAlpha Nuclide Database
File format	JSON
Character encoding	UTF-8
Default name	talpha_nuclides_<versionCode>.json
Compatibility	Additive extension of specification v2.2; backward-compatible with v1/v2 readers that ignore unknown fields

1. Purpose and compatibility

Version 3.0 extends specification v2.2 by adding adoptedLevels: a normalized list of evaluated adopted levels of the ordinary alpha-decay daughter nucleus that are energetically accessible from the selected parent state. The existing v1/v2 nuclide fields and alphaDecay content are preserved. Existing applications may ignore adoptedLevels, and a v3-compatible reader must accept older records in which the field is absent.

- The meanings and units of Z, symbol, massExcess, qAlpha and estimated are unchanged.
- Unknown data are represented by an absent optional field, not by a fabricated numerical value.
- qAlpha = 0 retains the v1 meaning: no Q-alpha value is supplied.

2. Top-level and layer structure

```
{"versionCode": 3, "formatVersion": "3.0", "layers": [{"A": 210, "nuclides": [...]}]}
```

Field	Type	Requirement
versionCode	integer	Required; update ordering, as in v1
formatVersion	string	Optional for v1/v2; recommended value "3.0" in v3.0

layers	array	Required
layers[].A	integer	Required
layers[].nuclides	array	Required

3. Nuclide record

```
{"Z":89,"symbol":"Ac","massExcess":...,"qAlpha":...,"estimated":false,"T_hl_exp":...,"alphaDecay"
:{...},"adoptedLevels":[...]}
```

Field	Type / unit	Meaning
Z	integer	Atomic number; required
symbol	string	Conventional element symbol; required
massExcess	number, keV	Atomic mass excess; required
qAlpha	number, keV	Ground-state to ground-state Q-alpha; required; 0 = unavailable
estimated	boolean	Status of massExcess only; required
T_hl_exp	number, years	Optional evaluated parent ground-state alpha-decay half-life, two significant figures
alphaDecay	object	Optional complete normalized ENSDF alpha-decay information
adoptedLevels	array	Optional normalized daughter adopted levels energetically accessible for alpha decay; empty array is allowed when the candidate-layer file is intentionally present but no levels are available

4. alphaDecay object

The object stores all alpha-decay information available in the imported ENSDF response. It is optional because ENSDF may contain no evaluated alpha-decay dataset for a nuclide.

```
"alphaDecay": {
  "decays": [{"decayId": 123, "decayMode": "A", "parent": {...},
    "daughter": {...}, "halfLife": {...}, "halfLifeSeconds": {...},
    "halfLifeYears": 1.1e+14, "qValue": {...}, "branchingRatio": {...},
    "normalizations": {...}, "dataset": {...}, "comments": [...]}],
  "transitions": [{...}]
}
```

Member	Meaning
decays	All alpha-decay edges for which the selected nuclide is explicitly the parent
transitions	All alpha-radiation records associated with the matched dataset IDs
parent / daughter	ENSDF identifiers, A, Z, state energy and spin/parity where available
halfLife	Original evaluated quantity and uncertainty in the ENSDF display unit
halfLifeSeconds	ENSDF canonical value in seconds
halfLifeYears	Derived seconds / 31,557,600; two significant figures
qValue	Evaluated decay Q value with unit and uncertainty
branchingRatio / normalizations	ENSDF normalization quantities without reinterpretation
dataset	ENSDF dataset ID, name, document type, database and cutoff date
comments	Evaluator comments, if requested from the API

5. Quantity and uncertainty representation

```
{"value": 2205.0, "unit": "keV", "uncertainty": {"type": "symmetric", "value": 1.0}}
```

Quantity objects are copied without deleting unfamiliar members. This preserves symmetric, asymmetric, limit, approximate and evaluator-entered forms supported by ENSDF. Importers must not convert a limit or an approximate value into an exact measurement.

6. Selection rules

- A record is an alpha decay only when decayMode is A/ALPHA (or the alpha character).
- The requested nuclide must match record.parent. A match in record.daughter is insufficient.

- Ground-state T_hl_exp is selected only from a parent state with levelEnergy = 0 keV (within the stored precision). Isomeric-state half-lives remain inside alphaDecay.decays.
- Alpha-transition records are joined through dataset ID; they must not be attached merely because A or Z happens to match.
- If several evaluated ground-state datasets remain, preserve all of them and leave scalar T_hl_exp absent until an explicit source-selection policy resolves the conflict.

7. Units and numerical format

Quantity	Canonical unit	Storage rule
massExcess	keV	Original database precision
qAlpha	keV	Original database precision
T_hl_exp	year	Scientific notation conceptually; JSON number; two significant figures
halfLifeSeconds	s	ENSDF canonical value
level/alpha energy	keV	Quantity object retains explicit unit
branching/intensity	dimensionless	Do not silently convert fraction to percent
adoptedLevels[].energy	keV	Quantity object retains explicit unit and uncertainty
adoptedLevels[].qAlphaLevel_keV	keV	Derived Q-alpha available to the corresponding daughter level
adoptedLevels[].l_min	dimensionless integer or null	Minimum orbital angular momentum when determinable; null when not determined

One year is defined as 365.25 days = 31,557,600 s. JSON does not preserve a textual exponent style; 1.1e+14 and 110000000000000 denote the same number. A serializer intended for human inspection should emit scientific notation.

8. ENSDF provenance

Every imported record should retain the ENSDF record ID and dataset metadata. The cutoffDate describes the evaluation, not the download date. For archival extraction, the converter output should additionally retain each matched original record under raw. The production mobile database may omit raw to reduce size after validation.

9. Validation

- A and Z are integers and identify the same nuclide as symbol.
- All required v1 fields exist and have the v1 types.
- Every alphaDecay.decays item has an alpha mode and the selected nuclide as parent.
- Daughter values satisfy A_d = A_p - 4 and Z_d = Z_p - 2 for ordinary alpha decay.
- T_hl_exp > 0 when present; qAlpha and adoptedLevels[].qAlphaLevel_keV are in keV; massExcess sign is physical. For every adopted level with a numeric excitation energy, qAlphaLevel_keV must equal the selected parent-to-daughter ground-state Q-alpha minus that excitation energy within stored precision.
- A failed or empty import never overwrites the last validated database.

10. Complete ENSDF archival format

This section defines the complete intermediate JSON written by get_alpha_data_from_ensdf(A, Z). It is an archival interchange file, not a replacement for the production talpha_nuclides_<versionCode>.json database. Its purpose is to preserve the complete API response needed for reproducible conversion, validation and later reprocessing.

```
{
  "source": {"database": "ENSDF", "api": "https://www.nndc.bnl.gov/ensdf-api"},
  "parent": {"A": 152, "Z": 64, "symbol": "Gd", "name": "152Gd"},
  "daughter": {"A": 148, "Z": 62, "symbol": "Sm", "name": "148Sm"},
  "alphaDecays": [...],
  "datasets": [...],
  "alphaRadiations": [...]
}
```

Member	Type	Meaning
source	object	Source database and API base address
parent	object	Requested parent nucleus: A, Z, symbol and display name
daughter	object	Ordinary alpha-decay daughter, calculated as A-4 and Z-2
alphaDecays	array	Complete matching records returned by /datasets/{id}/decays
datasets	array	Complete dataset objects returned by /datasets/{id}
alphaRadiations	array	Complete alpha-line records returned by /datasets/{id}/alphas

10.1 source object

Field	Type	Rule
database	string	Required; value ENSDF for this importer
api	string	Required; API base address used for acquisition

10.2 parent and daughter objects

Field	Type	Rule
A	integer	Mass number
Z	integer	Atomic number
symbol	string	Conventional element symbol
name	string	Compact nuclide name, for example 152Gd

For ordinary alpha decay the daughter object must satisfy daughter.A = parent.A - 4 and daughter.Z = parent.Z - 2. The object identifies the ENSDF nuclide under which the alpha-decay dataset is stored.

10.3 alphaDecays array

Every item is retained exactly as returned by the ENSDF dataset decay endpoint. Typical members include decayMode, parent, daughter, id and datasetSummary. Parent data may contain nuclideId, z, a, elementSymbol, name, spinParity, normalizations, levelEnergy, halfLife, halfLifeSeconds and qValue. Unknown and future ENSDF members must be preserved without reinterpretation.

10.4 datasets array

Every item is the complete metadata/content object for a matched alpha-decay dataset. Dataset identifiers are the authoritative join keys connecting alphaDecays, datasets and alphaRadiations. The importer must retain unfamiliar members so that the archive remains lossless across API revisions.

10.5 alphaRadiations array

Every item is a complete alpha-radiation record belonging to a matched dataset. Typical information includes alpha energy, uncertainty, intensity, hindrance factor, parent/daughter level references, comments and evaluator input where available. Records must be joined by dataset ID, never only by A or Z.

10.6 File naming and empty results

```
ENSDF_alpha_<A><Symbol>.json   Example: ENSDF_alpha_152Gd.json
```

- UTF-8 encoding and valid JSON are required.
- alphaDecays, datasets and alphaRadiations are always arrays, including when empty.
- An empty or failed acquisition must not overwrite a previously validated archive.
- The archive may be large and is not intended for direct loading by the Android application.

10.7 Mapping into the TAlpha database

ENSDF archive	TAlpha production record
alphaDecays	alphaDecay.decays after normalization
alphaRadiations	alphaDecay.transitions
parent.halfLifeSeconds	T_hl_exp after ground-state selection and conversion to years
parent.qValue	qAlpha after source policy and unit validation
datasets / datasetSummary	dataset metadata inside each normalized decay
complete raw records	Optional raw members for archival builds; may be omitted in mobile builds
daughter adopted-level dataset	adoptedLevels after normalization, energy filtering and derived qAlphaLevel_keV calculation

The archival file alone is not a complete nuclide record because massExcess and its estimated flag belong to the existing TAlpha mass-data layer. The deterministic stage-2 converter merges validated ENSDF alpha data into that existing record.

11. Important API query note

ENSDF alpha-decay datasets are stored under the daughter nuclide. For ordinary alpha decay, the importer calculates (A-4, Z-2), requests /nuclides/<daughter>/datasets, selects documentType = alphaDecay, then verifies locally that every retained decay record has the requested parent and decayMode A/ALPHA. A match in record.daughter is insufficient.

12. Recommended two-stage pipeline

- Stage 1: download and archive the complete ENSDF API response(s), including decays, matched datasets and alpha-radiation records.
- Stage 2: run a deterministic offline parser, validate parent/mode/dataset joins, obtain the daughter adopted-level dataset, calculate energetically accessible adoptedLevels, and generate the compact TAlpha database.
- Publish only after schema and physical-consistency validation; retain the archived input for reproducibility.

13. adoptedLevels extension (v3.0)

The adoptedLevels field stores evaluated levels of the ordinary alpha-decay daughter nucleus that can in principle be populated by alpha decay from the selected parent state. It is a candidate-level list, not a list of experimentally observed alpha branches. Experimentally measured or evaluated alpha transitions remain represented by alphaDecay.transitions. Thus, a level may occur in adoptedLevels without any corresponding observed alpha transition.

```
"adoptedLevels": [
  {
    "id": 131905,
    "datasetId": 5578,
    "levelIndex": 9,
    "isStable": false,
    "counts": {"feedingGammas": 0, "emittedGammas": 0},
    "spinParity": {
      "values": [{"twoTimesSpin": 2, "spin": 1.0, "relativeTo": {"label": "J"}}],
      "stringRepresentation": "J+1"
    },
    "offsetLabel": "Y",
    "energy": {
      "value": 140.4,
      "unit": "keV",
      "uncertainty": {"type": "symmetric", "value": 0.5}
    },
    "qAlphaLevel_keV": 7445.62,
    "l_min": null
  }
]
```

13.1 Semantics and relationship to alphaDecay

- adoptedLevels describes potential final states in the daughter nucleus; alphaDecay.transitions describes alpha transitions that are actually present in the evaluated alpha-decay dataset.
- The two arrays are complementary and must not be treated as interchangeable. Absence of a matching alphaDecay transition does not invalidate an adopted level.

- When a measured transition can be associated with an adopted level using ENSDF level identifiers, dataset references, or validated energy matching, the application or converter may use that association for display and comparison, but the original records remain preserved.
- Only energetically accessible levels are retained: qAlphaLevel_keV must be positive for an allowed candidate in the generated candidate list. A producer may retain the zero-threshold boundary only if explicitly required by its source-selection policy.

13.2 adoptedLevels item

Member	Type	Meaning / rule
id	integer	ENSDF level record identifier; preserve when supplied by the source
datasetId	integer	Identifier of the adopted-level dataset from which the level was taken
levelIndex	integer	Index/order of the level in the source adopted-level dataset
isStable	boolean	Source stability flag, preserved without reinterpretation
counts	object	Source counts such as feedingGammas and emittedGammas; preserve unfamiliar members
spinParity	object, optional	Evaluated spin/parity structure as supplied by the source, including stringRepresentation and detailed values
offsetLabel	string, optional	ENSDF offset/band label where present; preserve verbatim
energy	quantity object	Excitation energy of the daughter level; normally in keV, with original uncertainty representation
halflife	quantity object, optional	Evaluated half-life of the

		daughter level when available; this is a level property, not the parent alpha-decay half-life
qAlphaLevel_keV	number, keV	Derived energy release available for alpha decay to this level: Q_alpha(level) = Q_alpha(gs→gs) - E_exc(level)
l_min	integer or null	Minimum alpha-particle orbital angular momentum required by the adopted parent/daughter spin-parity assignments when uniquely determinable; null when it cannot be determined reliably

13.3 Construction rules

- The daughter nucleus is the ordinary alpha-decay daughter: A_d = A_p - 4 and Z_d = Z_p - 2.
- Levels are taken from the evaluated adopted-level dataset selected for that daughter nucleus. Source identifiers and unfamiliar source members should be preserved whenever practical.
- For a level with excitation energy E_exc, calculate qAlphaLevel_keV = qAlphaTheor_keV - E_exc when qAlphaTheor_keV is the selected theoretical/working ground-state Q-alpha used by the candidate generator. If the production database uses qAlpha as the selected ground-state Q-alpha, the same relation applies using qAlpha.
- Retain only levels for which the calculated qAlphaLevel_keV is energetically accessible according to the candidate-generation policy.
- Duplicate source levels must not be merged solely because their numerical energies are equal. Distinct level IDs, levelIndex values, offset labels, or band assignments may represent physically distinct evaluated levels.
- l_min is derived only when the available spin/parity information permits an unambiguous minimum orbital angular momentum under alpha-decay angular-momentum and parity conservation. Otherwise store null rather than inventing a value.

13.4 Empty and unavailable data

- adoptedLevels is optional in older v1/v2 production records.
- For a v3 candidate-output file that is intentionally created for a nuclide, adoptedLevels should be written as an array even when no adopted levels are available; use "adoptedLevels": [].

- A missing or empty adoptedLevels array does not mean that alpha decay is experimentally excluded. It means that no candidate daughter-level records are supplied by this layer.
- Likewise, alphaDecay may be absent or may contain no measured transitions even when adoptedLevels contains energetically accessible candidate levels.

13.5 Example candidate-level file

```
{
  "A": 210,
  "Z": 89,
  "symbol": "Ac",
  "daughter": {"A": 206, "Z": 87, "symbol": "Fr"},
  "qAlphaTheor_keV": 7586.02,
  "adoptedLevels": [
    {
      "id": 131905,
      "datasetId": 5578,
      "levelIndex": 9,
      "energy": {"value": 140.4, "unit": "keV",
                 "uncertainty": {"type": "symmetric", "value": 0.5}},
      "spinParity": {"stringRepresentation": "J+1", "values": [...]},
      "qAlphaLevel_keV": 7445.62,
      "l_min": null
    }
  ]
}
```

The candidate-level file shown above is an intermediate/derived representation used by the database-building pipeline. The production TAlpha nuclide record may embed the adoptedLevels array directly while retaining the existing v1/v2 fields and optional alphaDecay object.